

CMU MSCF Program - Special Lectures

From the Binomial Model to Black-Scholes

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Page 1: The One-Period Binomial Model

- ▶ A probability space $(\Omega, \mathbb{P})$, with $\Omega = \{H, T\}$ and $\mathbb{P}(H) > 0$, $\mathbb{P}(T) > 0$. (Think of H and T representing “head” and “tail” in a coin toss. The coin need not be “fair”.)
- ▶ Two investment times: $t = 0$ and $t = 1$
- ▶ A bank at which one can borrow or invest between $t = 0$ and $t = 1$ at the one-period interest rate $r > -1$
- ▶ A single stock S (no dividends) whose price at time 0 is S_0 . The price S_1 of the stock at time 1 is a random variable on Ω satisfying $S_1(H) = uS_0$ and $S_1(T) = dS_0$. Here u and d are given real numbers with $u > d > 0$.

Page 2: Strategies

DEFINITION: A **strategy** (or **portfolio**) is described by a pair (X_0, Δ_0) of real numbers, where X_0 is the initial capital and Δ_0 is the number of shares of stock held. (The amount of money in the bank at time 0 is $X_0 - \Delta_0 S_0$.) The terminal capital of a strategy is given by

$$X_1(\omega) = (X_0 - \Delta_0 S_0)(1 + r) + \Delta_0 S_1(\omega) \text{ for all } \omega \in \{H, T\}.$$

DEFINITION: An **arbitrage strategy** is a strategy with

- (a) $X_0 = 0$,
- (b) $\mathbb{P}[X_1 \geq 0] = 1$, and
- (c) $\mathbb{P}[X_1 > 0] > 0$.

Page 3: No-Arbitrage Condition

PROPOSITION: The one-period binomial model is free of arbitrage if and only if

$$d < 1 + r < u.$$

The proof is quite straightforward. (If $d \geq 1 + r$, one can create an arbitrage by borrowing S_0 from the bank and buying one share of stock. If $u \leq 1 + r$, one can create an arbitrage by selling short one share of stock and investing S_0 in the bank. For $d < 1 + r < u$ it is straightforward to check that if $X_0 = 0$ and $\mathbb{P}[X_1 \geq 0] = 1$, then $\mathbb{P}[X_1 > 0] = 0$.)

Page 4: Derivative Securities

By a **derivative security** or **contingent claim** in the one-period binomial model we mean a security V with payoff at time 1 a random variable V_1 on Ω .

EXAMPLES:

- ▶ Zero Coupon Bond with Face Value F : $V_1(\omega) = F$
- ▶ Stock: $V_1(\omega) = S_1(\omega)$
- ▶ Call Option with strike K : $V_1(\omega) = (S_1(\omega) - K)^+$
- ▶ Put Option with strike K : $V_1(\omega) = (K - S_1(\omega))^+$

Recall that

$$x^+ = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

Page 5: Replication and Pricing

Consider a security V with payment function $V_1 : \Omega \rightarrow \mathbb{R}$. In order to replicate this security, we need to find a strategy (X_0, Δ_0) such that

$$X_1(\omega) = V_1(\omega) \text{ for all } \omega \in \{H, T\}.$$

This is equivalent to the pair of equations

$$\Delta_0 u S_0 + (1 + r)(X_0 - \Delta_0 S_0) = V_1(H)$$

$$\Delta_0 d S_0 + (1 + r)(X_0 - \Delta_0 S_0) = V_1(T).$$

This pair of equations always has exactly one solution.

Page 6: Replication and Pricing (Cont.)

Subtracting the equations, we find

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{uS_0 - dS_0} = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}.$$

Substituting Δ_0 into either equation gives

$$X_0(1+r) = \left(\frac{1+r-d}{u-d} \right) V_1(H) + \left(\frac{u-1-r}{u-d} \right) V_1(T).$$

It is convenient to define

$$\tilde{p} = \frac{1+r-d}{u-d}, \quad \tilde{q} = \frac{u-1-r}{u-d}.$$

Notice that $\tilde{p} + \tilde{q} = 1$ and that the no-arbitrage condition is equivalent to $\tilde{p}, \tilde{q} > 0$.

Page 7: Replication and Pricing (Cont.)

The formula for X_0 becomes

$$X_0 = \frac{1}{1+r}(\tilde{p}V_1(H) + \tilde{q}V_1(T)).$$

We conclude that V is replicable and that the arbitrage-free price V_0 is given by

$$V_0 = \frac{1}{1+r}(\tilde{p}V_1(H) + \tilde{q}V_1(T)).$$

PRICING = REPLICATION

Page 8: Pricing Measure aka Risk-Neutral Measure

Define $\tilde{\mathbb{P}} : \Omega \rightarrow \mathbb{R}$ by

$$\tilde{\mathbb{P}}(H) = \tilde{p}, \quad \tilde{\mathbb{P}}(T) = \tilde{q},$$

and observe that $\tilde{\mathbb{P}}$ is a probability measure on Ω . It is called a **pricing measure** or **risk-neutral measure**. The numbers $\tilde{p}$ and $\tilde{q}$ are called **pricing probabilities** or **risk-neutral probabilities**. They do not represent probabilities of actual events occurring in the model.

REMARK: The term “risk-neutral measure” is used much more commonly than “pricing measure”. However, the term “risk-neutral” seems to lead to a lot of confusion with students when they first encounter these ideas. **David Heath** (co-inventor of the HJM model and former colleague) strongly advocated using the language “pricing measure” and “pricing probabilities” when introducing this material to students.

Page 9: Pricing Measure (Cont.)

The arbitrage-free price of a derivative security V can be expressed in the form

$$V_0 = \frac{1}{1+r} \tilde{\mathbb{E}}[V_1],$$

where $\tilde{\mathbb{E}}$ denotes expected value under the measure $\tilde{\mathbb{P}}$.

THE ARBITRAGE-FREE PRICE OF A DERIVATIVE SECURITY IS EQUAL TO THE DISCOUNTED RISK-NEUTRAL EXPECTED VALUE OF THE PAYOFF.

Page 10: Pricing Measure (Cont.)

REMARK: There is no need to memorize the formulas for $\tilde{p}$ and $\tilde{q}$ because they follow easily from observing that $\tilde{\mathbb{E}}[S_1] = (1+r)S_0$:

$$S_0(1+r) = \tilde{p}uS_0 + \tilde{q}dS_0,$$

$$\tilde{p} + \tilde{q} = 1.$$

Just solve the equations!

Page 11: Example: Pricing a Put and a Call

Consider a one-period binomial model with $S_0 = 50$, $u = 2$, $d = .5$, and $r = .25$. (Notice that $S_1(H) = 100$ and $S_1(T) = 25$.)

Observe that the pricing probabilities are given by

$$\tilde{p} = \frac{1 + .25 - .5}{2 - .5} = .5, \quad \tilde{q} = 1 - \tilde{p} = .5.$$

Let P be a put option with $K = 40$ so that $P_1(H) = 0$ and $P_1(T) = 15$.

It follows that

$$P_0 = \frac{1}{1.25}(0 + .5(15)) = 6.$$

Page 12: Example (Cont.)

The number of stock shares in the replicating portfolio is

$$\Delta_0 = \frac{P_1(H) - P_1(T)}{S_1(H) - S_1(T)} = \frac{0 - 15}{100 - 25} = -\frac{1}{5}.$$

Let C denote a call with strike $K = 40$, and observe that $C_1(H) = 60$, $C_1(T) = 0$.

It follows that

$$C_0 = \frac{1}{1.25}(.5(60) + 0) = 24,$$

and the number of stock shares in the replicating portfolio is

$$\Delta_0 = \frac{C_1(H) - C_1(T)}{S_1(H) - S_1(T)} = \frac{60 - 0}{100 - 25} = \frac{4}{5}.$$

Page 13: Multi-Period Binomial Model

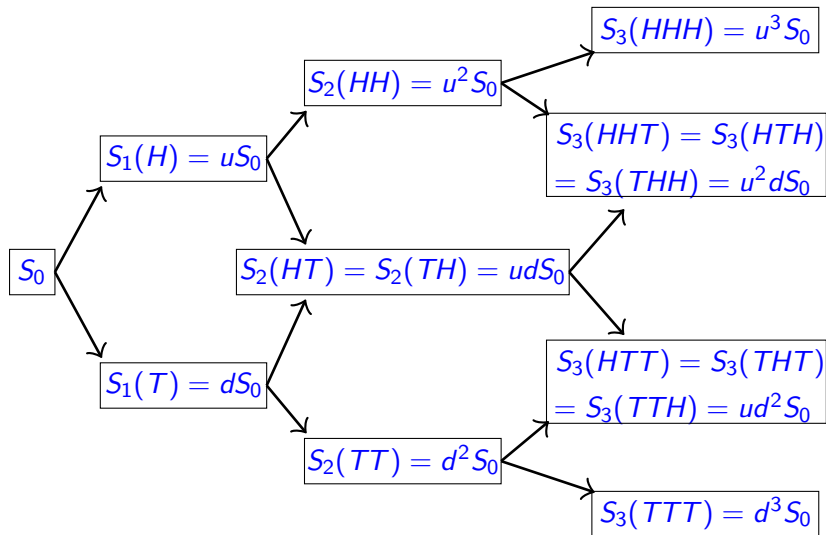
- ▶ $N + 1$ trading times, $t = 0, 1, \dots, N$. (N investment periods.)
- ▶ A probability space $(\Omega, \mathbb{P})$, where $\Omega = \{H, T\}^N$, and $\mathbb{P}(\omega) > 0$ for all $\omega \in \Omega$. ($\mathbb{P}$ is called the **actual measure**, **objective measure**, or **physical measure**.)
- ▶ A bank at which one can borrow or invest at the one-period interest rate $r > -1$.
- ▶ A single stock S (that pays no dividends). The initial price (per share) of the stock is S_0 . There are numbers $u > d > 0$ such that the price S_{n+1} of the stock at time $n + 1$ is related to the price S_n of the stock at time n by

$$S_{n+1}(\omega) = \begin{cases} uS_n(\omega) & \text{if } \omega_{n+1} = H \\ dS_n(\omega) & \text{if } \omega_{n+1} = T. \end{cases}$$

“Tree” of One-Period Binomial Models

Page 14: Three-Period Tree

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$



Page 15: Adapted Processes and Information

As time evolves, more information becomes available in financial markets. Financial decisions that are made at time n must be based only the information that is available at time n . *You can't have tomorrow's Wall Street Journal today!* The mathematical concept that captures this idea is the notion of an *adapted process*. In the binomial model the information that is available at time n is the outcomes of the first n coin tosses.

DEFINITION: Let $m \in \{0, 1, \dots, N\}$ be given. A list $(W_n)_{0 \leq n \leq m}$ of random variables on Ω is said to be an *adapted process* provided that for every $n \in \{0, 1, 2, \dots, m\}$ and every $\omega \in \Omega$ the value of $W_n(\omega)$ is completely determined by $(\omega_1, \omega_2, \dots, \omega_n)$. (In other words, the value of W_n is completely determined by the outcomes of the first n coin tosses – W_n becomes known at time n .)

Notice that $(S_n)_{0 \leq n \leq N}$ is an adapted process.

Page 16: Self-Financing Strategies

Since there is more than one investment period, we need to worry about whether capital could be introduced (or removed) at intermediate times. A strategy is said to be **self-financing** provided that capital is introduced (or removed) only at the initial time. (Of course, assets can be reallocated at each investment time.)

A self-financing strategy in the N -period binomial model is described by an initial capital X_0 and an adapted process $(\Delta_n)_{0 \leq n \leq N-1}$. The random variable Δ_n represents the number of shares of stock held between time n and time $n+1$. (The requirement that the process be adapted reflects the idea that strategies should not be able to “look into the future”.)

The capitals in a self-financing strategy evolve according to

$$X_{n+1} = (X_n - \Delta_n S_n)(1 + r) + \Delta_n S_{n+1}.$$

Page 17: Arbitrage

DEFINITION: By an **arbitrage strategy** we mean a self-financing strategy satisfying

- (a) $X_0 = 0$,
- (b) $\mathbb{P}[X_N \geq 0] = 1$,
- (c) $\mathbb{P}[X_N > 0] > 0$.

THEOREM: The multi-period binomial model is free of arbitrage if and only if

$$d < 1 + r < u.$$

From now on, we assume that $d < 1 + r < u$.

Page 18: Backward Induction

The multi-period binomial model can be viewed as a tree of one-period models. It is therefore convenient to introduce the **one-step pricing probabilities** or **one-step risk-neutral probabilities** by

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - 1 - r}{u - d}.$$

It is straightforward to prove the following proposition.

Page 19: Backward Induction (Cont.)

PROPOSITION: An adapted process $(X_n)_{0 \leq n \leq N}$ is the capital of a self-financing strategy if and only if for every $n \in \{0, 1, \dots, N-1\}$ and every $\omega_1, \omega_2, \dots, \omega_n \in \{H, T\}$ we have

$$X_n(w_n) = \frac{1}{1+r} [\tilde{p}X_{n+1}(w_n, H) + \tilde{q}X_{n+1}(w_n, T)],$$

where $w_n = (\omega_1, \omega_2, \dots, \omega_n)$. In this case we have

$$\Delta_n(w_n) = \frac{X_{n+1}(w_n, H) - X_{n+1}(w_n, T)}{S_{n+1}(w_n, H) - S_{n+1}(w_n, T)}.$$

Page 20: European-Style Derivative Securities & Replication

DEFINITION: By a **European-style derivative security** (with maturity N) we mean a security V that makes a single payment of amount V_N at time N , where V_N is a random variable on Ω .

DEFINITION: By a **replicating strategy** for a European-style derivative security having payment function V_N we mean a self-financing strategy X such that

$$X_N(\omega) = V_N(\omega) \text{ for all } \omega \in \Omega.$$

Page 21: Replication (Cont.)

THEOREM: Let V be a European-style derivative security. Then there is exactly one replicating strategy. This strategy can be constructed by backward induction as follows:

- ▶ $X_N(\omega) = V_N(\omega)$ for all $\omega \in \Omega$.
- ▶ Let $n \in \{0, 1, \dots, N-1\}$ be given. Assuming that X_{n+1} has already been defined, we define X_n and Δ_n by

$$X_n(w_n) = \frac{1}{1+r} [\tilde{p}X_{n+1}(w_n, H) + \tilde{q}X_{n+1}(w_n, T)],$$

$$\Delta_n(w_n) = \frac{X_{n+1}(w_n, H) - X_{n+1}(w_n, T)}{S_{n+1}(w_n, H) - S_{n+1}(w_n, T)},$$

with $w_n = (\omega_1, \omega_2, \dots, \omega_n)$.

The **arbitrage-free price** of a European-style derivative security V is equal to the initial capital of the replicating strategy.

Page 22: The Rollback Operator

The backward induction formula for the capitals can be written more compactly as

$$X_n = \mathcal{R}_n[X_{n+1}],$$

where the **rollback operator** $\mathcal{R}_n$ is defined by

$$\mathcal{R}_n[X_{n+1}](w_n) = \frac{1}{1+r} [\tilde{p}X_{n+1}(w_n, H) + \tilde{q}X_{n+1}(w_n, T)],$$

with $w_n = (\omega_1, \omega_2, \dots, \omega_n)$.

To find the arbitrage-free price V_0 of a European-style derivative security V , one can use the algorithm

- ▶ $X_N = V_N$,
- ▶ $X_n = \mathcal{R}_n[X_{n+1}]$ for $n = N-1, N-2, \dots, 0$,
- ▶ $V_0 = X_0$.

Page 23: Pricing Measure

The backward induction algorithm can be used to express V_0 as a sum of 2^N terms. Each term will have a factor $\frac{1}{(1+r)^N}$ and factors $\tilde{p}^{\#H(\omega)}$ and $\tilde{q}^{\#T(\omega)}$ multiplying $V(\omega)$. Here, $\#H(\omega)$ and $\#T(\omega)$ are the number of heads and tails in the string ω

This suggests that we define the measure $\tilde{\mathbb{P}}$ on Ω by

$$\tilde{\mathbb{P}}(\omega) = \tilde{p}^{\#H(\omega)} \tilde{q}^{\#T(\omega)},$$

so that the pricing formula becomes

$$V_0 = \frac{1}{(1+r)^N} \sum_{\omega \in \Omega} \tilde{\mathbb{P}}(\omega) V_N(\omega) = \frac{1}{(1+r)^N} \tilde{\mathbb{E}}[V_N].$$

Page 24: Pricing Measure (Cont.)

The measure $\tilde{\mathbb{P}}$ is called the **pricing measure**, **risk-neutral measure**, or **martingale measure**.

Notice that the coin tosses are independent under $\tilde{\mathbb{P}}$. (We did not assume independence of the coin tosses under the actual measure.)

So far the actual measure has not been important for anything. (We have used only the property that every path can actually occur.) The actual measure will play a crucial role in optimal investment.

Page 25: Conditional Expectations and Martingales

Let Y be a random variable on Ω whose value is completely determined by the outcomes of the first $n + 1$ coin tosses. Then the conditional expectation at time n of Y under $\tilde{\mathbb{P}}$ is given by

$$\tilde{\mathbb{E}}_n[Y](w_n) = \tilde{p}Y(w_n, H) + \tilde{q}Y(w_n, T),$$

where $w_n = (\omega_1, \omega_2, \dots, \omega_n)$. [This is not the real definition of conditional expectation – only a special case – you will see a true definition in Stochastic Calculus.]

DEFINITION: An adapted process $(M_n)_{0 \leq n \leq N}$ is said to be a **martingale** under $\tilde{\mathbb{P}}$ provided that

$$M_n = \tilde{\mathbb{E}}_n[M_{n+1}] \quad \text{for all } n = 0, 1, \dots, N - 1.$$

Page 26: Conditional Expectations & Martingales (Cont.)

The backward induction algorithm for capitals of a self-financing strategy can be expressed as

$$X_n = \frac{1}{1+r} \tilde{\mathbb{E}}_n[X_{n+1}].$$

It follows easily that the discounted capitals

$$\left(\frac{X_n}{(1+r)^n} \right)_{0 \leq n \leq N}$$

form a martingale under $\tilde{\mathbb{P}}$.

Page 27: Martingales (Cont.)

Martingales have many “magical properties”. The theory of martingales can be used to lend great efficiency and elegance to the study of multi-period binomial models.

Although the multi-period model can be completely analyzed without the use of martingales, it is extremely beneficial to use introduce the martingale approach. In addition to gaining elegance and efficiency, the discrete setting is an excellent vehicle for becoming comfortable with martingales. Martingales are essential (but more difficult to get a feel for) in the continuous time setting.

Page 28: The Black-Scholes Formulas

The so-called Black-Scholes formulas were published by Fisher Black and Myron Scholes in 1973. Many people found their arguments difficult to understand and objected to their conclusions because the put and call prices did not depend explicitly on the expected return of the underlying stock. Important and substantial clarifications were provided by Robert Merton. Merton and Scholes shared the Nobel prize in Economics in 1997. (Fisher Black died in 1995.)

Page 29: Use of the Binomial Model

The binomial model was introduced by Cox, Ross, and Rubinstein in 1979. I believe that their primary motivation for inventing the model was to provide a derivation of the Black-Scholes formulas that is easier to understand.

The approach presented here is a bit different than that of Cox, Ross, and Rubinstein, and is based substantially on some lectures given by David Heath in 2001.

Page 30: Indexing

Let us fix a time T (measured in years). We shall divide the time interval $[0, T]$ into N steps, so that each step corresponds to a time period of length $\frac{T}{N}$ in years. We want to build N -period binomial models (appropriately scaled) and let $N \rightarrow \infty$.

We need to keep track of how many periods there are in the model as well as the time step within the model. We use an upper index (in parentheses) to indicate the number N of periods in the model, and a lower index to indicate the time period n within the N -period model, i.e. $S_{N-1}^{(N)}$ denotes the stock price one-period before the end when there are N periods in the model.

In particular, $S_{N-1}^{(N)}$ corresponds to the stock price in

$$\left(\frac{N-1}{N} \right) T \text{ years.}$$

Page 31: Actual Measure and Model Parameters

SAMPLE SPACE: $\Omega^{(N)} = \{H, T\}^N$

We'll take the actual measure to correspond independent tosses of identical coins, having probability of a head equal to $\frac{1}{2}$, i.e.

$$\mathbb{P}^{(N)}(\omega) = \left(\frac{1}{2}\right)^N \quad \text{for all } \omega \in \Omega^{(N)}.$$

We denote the parameters in the N -period model by $u^{(N)}$, $d^{(N)}$, and $r^{(N)}$. These must scale with N in suitable ways in order to obtain a meaningful limit.

The initial stock price will be independent of N , i.e.

$$S_0^{(N)} = S_0 \quad \text{for all } N \in \mathbb{N}.$$

Page 32: One-Period Stock Returns

The return of the stock during the period from $n - 1$ to n is the random variable $\rho_n^{(N)}$ on $\Omega^{(N)}$ defined by

$$\rho_n^{(N)} = \frac{S_n^{(N)} - S_{n-1}^{(N)}}{S_{n-1}^{(N)}} = \frac{S_n^{(N)}}{S_{n-1}^{(N)}} - 1.$$

Under $\mathbb{P}^{(N)}$, the random variables $\rho_n^{(N)}$, $n = 1, 2, \dots, N$ are independent and identically distributed. In fact, we have

$$\rho_n^{(N)} = \begin{cases} u^{(N)} - 1 & \text{with probability } \frac{1}{2} \\ d^{(N)} - 1 & \text{with probability } \frac{1}{2}. \end{cases}$$

Let us denote their common mean by $\alpha^{(N)}$ and standard deviation by $\sigma^{(N)}$.

Page 33: One-Period Stock Returns (Cont.)

A simple computation gives

$$\alpha^{(N)} = \mathbb{E}^{(N)}[\rho_n^{(N)}] = \frac{1}{2}(u^{(N)} + d^{(N)}) - 1,$$

$$\sigma^{(N)} = \sqrt{\text{Var}^{(N)}[\rho_n^{(N)}]} = \frac{u^{(N)} - d^{(N)}}{2}.$$

Page 34: Form of $u^{(N)}$ and $d^{(N)}$

It follows that

$$u^{(N)} = 1 + \alpha^{(N)} + \sigma^{(N)},$$

$$d^{(N)} = 1 + \alpha^{(N)} - \sigma^{(N)}.$$

Of course, as N increases, the values of $\alpha^{(N)}$, $\sigma^{(N)}$, and $r^{(N)}$ must change, because each period represents a shorter time interval. In fact, we must have

$$\alpha^{(N)}, \sigma^{(N)}, r^{(N)} \rightarrow 0 \text{ as } N \rightarrow \infty.$$

The issue is to determine the precise rates of decay as $N \rightarrow \infty$.

Page 35: Little “Oh” Notation

If f and g are real-valued functions it is customary to write

$$f(x) = o(g(x)) \text{ as } x \rightarrow a$$

to indicate

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = 0,$$

and to write

$$f(N) = o(g(N)) \text{ as } N \rightarrow \infty$$

to indicate

$$\lim_{N \rightarrow \infty} \left(\frac{f(N)}{g(N)} \right) = 0.$$

For example,

$$x^4 = o(x^3) \text{ as } x \rightarrow 0,$$

and

$$\frac{1}{N^{\frac{3}{2}}} = o\left(\frac{1}{N}\right) \text{ as } N \rightarrow \infty.$$

Page 36: Approximations for $\ln(1+x)$ as $x \rightarrow 0$

Taylor Series about 0:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

The above expression is valid for $|x| < 1$. By truncating the series, we see that

$$\ln(1+x) = x + o(x) \text{ as } x \rightarrow 0,$$

$$\ln(1+x) = x - \frac{x^2}{2} + o(x^2) \text{ as } x \rightarrow 0.$$

The use of the little-oh notation will allow us to focus on the important terms by combining many unimportant terms together as a single “little-oh” term.

Page 37: Scaling the Interest Rate

An investment of \$1 in the bank at time 0 will be worth

$$(1 + r^{(N)})^N$$

after N periods.

We would like this quantity to have a finite and strictly positive limit as $N \rightarrow \infty$. To achieve this, we should have

$$(1) \quad r^{(N)} = \frac{c}{N} + o\left(\frac{1}{N}\right),$$

for some constant c . To see why this is the case, notice that

$$\ln[(1 + r^{(N)})^N] = N \ln(1 + r^{(N)}) = N[r^{(N)} + o(r^{(N)})] \text{ as } N \rightarrow \infty.$$

The above quantity will have a finite limit as $N \rightarrow \infty$ if and only if (1) holds.

Page 38: Scaling the Interest Rate (Cont.)

It is natural to take

$$(2) \quad r^{(N)} = \frac{rT}{N},$$

so that

$$\lim_{n \rightarrow \infty} (1 + r^{(N)})^N = e^{rT}.$$

In this case, r represents an annualized interest rate corresponding to continuous compounding.

Page 39: Scaling of $\alpha^{(N)}$ and $\sigma^{(N)}$

To understand how $\alpha^{(N)}$ and $\sigma^{(N)}$ should be scaled, we introduce the **total return** $R^{(N)}$ of the stock defined by

$$R^{(N)} = \frac{S_N^{(N)} - S_0^{(N)}}{S_0^{(N)}} = \frac{S_N^{(N)}}{S_0^{(N)}} - 1.$$

It is natural to require that the mean and variance of $R^{(N)}$ under $\mathbb{P}^{(N)}$ have finite limits as $N \rightarrow \infty$.

In order to compute the mean and variance of $R^{(N)}$, it is helpful to notice that

$$\begin{aligned} 1 + R^{(N)} &= \frac{S_N^{(N)}}{S_{N-1}^{(N)}} \frac{S_{N-1}^{(N)}}{S_{N-2}^{(N)}} \cdots \frac{S_1^{(N)}}{S_0^{(N)}} \\ &= (1 + \rho_N^{(N)})(1 + \rho_{N-1}^{(N)}) \cdots (1 + \rho_1^{(N)}). \end{aligned}$$

Page 40: Scaling (Cont.)

Using the independence of $\rho_n^{(N)}$ we find that

$$\mathbb{E}^{(N)}[1 + R^{(N)}] = (1 + \alpha^{(N)})^N.$$

In order for $\mathbb{E}^{(N)}[R^{(N)}]$ to have a finite limit as $N \rightarrow \infty$ we want to have

$$(3) \quad \alpha^{(N)} = \frac{c}{N} + o\left(\frac{1}{N}\right),$$

for some constant c . The reasoning is the same as with the interest rate.

Page 41: Scaling (Cont.)

In order for the constant to have an “annualized” interpretation, we take

$$(4) \quad \alpha^{(N)} = \frac{\alpha T}{N},$$

for a constant α , which we call the **continuously compounded expected annual return**.

Page 42: Scaling (Cont.)

To compute $\text{Var}^{(N)}[R^{(N)}]$ we observe that

$$\begin{aligned}\mathbb{E}^{(N)} \left[\left(\frac{S_n^{(N)}}{S_{n-1}^{(N)}} \right)^2 \right] &= \frac{1}{2}[(u^{(N)})^2 + (d^{(N)})^2] \\ &= (1 + \alpha^{(N)})^2 + (\sigma^{(N)})^2.\end{aligned}$$

Using the independence of the $\rho_n^{(N)}$, the formula

$$\text{Var}[Y] = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2,$$

and the fact that $\text{Var}[Y] = \text{Var}[1 + Y]$, we find that

$$\text{Var}^{(N)}[R^{(N)}] = [(1 + \alpha^{(N)})^2 + (\sigma^{(N)})^2]^N - (1 + \alpha^{(N)})^{2N}.$$

Page 43: Scaling (Cont.)

Substituting in

$$\alpha^{(N)} = \frac{\alpha T}{N},$$

we find that

$$\text{Var}^{(N)}[R^{(N)}] = \left[1 + \frac{2\alpha T}{N} + \frac{\alpha^2 T^2}{N^2} + (\sigma^{(N)})^2 \right]^N - \left(1 + \frac{\alpha T}{N} \right)^{2N}.$$

As $N \rightarrow \infty$,

$$\left(1 + \frac{\alpha T}{N} \right)^{2N} \rightarrow e^{2\alpha T}.$$

Page 44: Scaling (Cont.)

In order for $\text{Var}^{(N)}[R^{(N)}]$ to have a finite limit as $N \rightarrow \infty$, we want

$$\ln \left(\left[1 + \frac{2\alpha T}{N} + \frac{\alpha^2 T^2}{N^2} + (\sigma^{(N)})^2 \right]^N \right)$$

to have an finite limit as $N \rightarrow \infty$. Using the approximation $\ln(1+x) = x + o(x)$ as $x \rightarrow 0$, and the fact that $\ln(A^N) = N \ln A$, we conclude that

$$2\alpha T + N(\sigma^{(N)})^2$$

should have a finite limit as $N \rightarrow \infty$.

Page 45: Scaling (Cont.)

In other words, we need to have

$$(5) \quad \sigma^{(N)} = \frac{c}{\sqrt{N}} + o\left(\frac{1}{\sqrt{N}}\right).$$

Here we need to have $c \neq 0$; otherwise the limiting variance will be 0 and the randomness will be gone.

Again, we want the parameter that we introduce to have an “annualized” interpretation. We take

$$(6) \quad \sigma^{(N)} = \frac{\sigma\sqrt{T}}{\sqrt{N}},$$

for a constant $\sigma > 0$ called the (annualized) Black-Scholes volatility parameter.

Page 46: Risk-Neutral Measure

With the above scalings (2),(4), and (6), we have

$$u^{(N)} = 1 + \frac{\alpha T}{N} + \frac{\sigma\sqrt{T}}{\sqrt{N}}, \quad d^{(N)} = 1 + \frac{\alpha T}{N} - \frac{\sigma\sqrt{T}}{\sqrt{N}},$$

and

$$d^{(N)} < 1 + r^{(N)} < u^{(N)} \quad \text{for } N \text{ sufficiently large.}$$

The one-step risk-neutral probabilities are given by

$$\tilde{p}^{(N)} = \frac{1}{2} - \frac{(\alpha - r)\sqrt{T}}{2\sigma\sqrt{N}}, \quad \tilde{q}^{(N)} = \frac{1}{2} + \frac{(\alpha - r)\sqrt{T}}{2\sigma\sqrt{N}}.$$

Page 47: Limiting Distributions of the Stock Price

We want to understand how the distributions of $S_N^{(N)}$ under $\mathbb{P}^{(N)}$ and $\tilde{\mathbb{P}}^{(N)}$ behave when N is large. For pricing options, the distribution under $\tilde{\mathbb{P}}^{(N)}$ is what matters. Recall that

$$\frac{S_N^{(N)}}{S_0} = (1 + \rho_1^{(N)})(1 + \rho_2^{(N)}) \cdots (1 + \rho_N^{(N)}).$$

It is convenient to take logs. If we put

$$g_n^{(N)} = \ln(1 + \rho_n^{(N)}),$$

we get

$$\ln \left(\frac{S_N^{(N)}}{S_0} \right) = \sum_{n=1}^N g_n^{(N)}.$$

Page 48: Limiting Distributions (Cont.)

We focus on the distribution under $\tilde{\mathbb{P}}^{(N)}$ here. If we define

$$\zeta^{(N)} = \frac{(\alpha - r)\sqrt{T}}{\sigma\sqrt{N}},$$

then we can write

$$\tilde{\mathbb{P}}^{(N)} = \left(\frac{1}{2}\right)^N (1 - \zeta^{(N)})^{\#H(\omega)} (1 + \zeta^{(N)})^{\#T(\omega)}.$$

Page 49: Limiting Distributions (Cont.)

Under $\tilde{\mathbb{P}}^{(N)}$ the $g_n^{(N)}$, $n = 1, 2, \dots, N$ are independent and identically distributed. To obtain useful expressions for the mean and variance of these random variables, we use a 2-term Taylor expansion for the log:

$$\ln(1+x) = x - \frac{x^2}{2} + o(x^2).$$

We find that

$$\tilde{\mathbb{E}}^{(N)}[g_n^{(N)}] = \frac{\alpha T}{N} - \frac{\sigma^2 T}{2N} - \zeta^{(N)} \left(\frac{\sigma \sqrt{T}}{\sqrt{N}} \right) + o\left(\frac{1}{N}\right),$$

$$\tilde{\text{Var}}^{(N)}[g_n^{(N)}] = \frac{\sigma^2 T}{N} + o\left(\frac{1}{N}\right).$$

Page 50: Limiting Distributions (Cont.)

Summing up, we get

$$\tilde{\mathbb{E}}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] = \alpha T - \frac{1}{2} \sigma^2 T - \sigma \sqrt{T} \sqrt{N} \zeta^{(N)} + o(1)$$

$$\tilde{\text{Var}}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] = \sigma^2 T + o(1).$$

Remark: Observe that if we replace $\zeta^{(N)}$ with 0 then the formulas above would give us the expected value and variance of $S_N^{(N)}/S_0$ under the actual measure. This means that the term $\sigma \sqrt{T} \sqrt{N} \zeta^{(N)}$ appearing in the first equation is of real interest here because it will tell us what the difference is between the actual and risk-neutral distributions of the stock price.

Page 51: Limiting Distributions (Cont.)

Recall that

$$\zeta^{(N)} = \frac{(\alpha - r)\sqrt{T}}{\sigma\sqrt{N}},$$

so that

$$\sigma\sqrt{T}\sqrt{N}\zeta^{(N)} = (\alpha - r)T.$$

We conclude that

Page 52: Limiting Distributions (Cont.)

$$\mathbb{E}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] \approx \left(\alpha - \frac{\sigma^2}{2} \right) T,$$

$$\tilde{\mathbb{E}}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] \approx \left(r - \frac{\sigma^2}{2} \right) T,$$

We also have

$$\text{Var}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] \approx \sigma^2 T,$$

$$\tilde{\text{Var}}^{(N)} \left[\ln \left(\frac{S_N^{(N)}}{S_0} \right) \right] \approx \sigma^2 T.$$

Page 53: Limiting Distribution (Cont.)

In view of the Central Limit Theorem, when N is very large the distribution of $\ln(S_N^{(N)}/S_0)$

- ▶ under the actual measure will be nearly normal with mean $(\alpha - \frac{\sigma^2}{2})T$ and variance $\sigma^2 T$.
- ▶ under the risk-neutral measure will be nearly normal with mean $(r - \frac{\sigma^2}{2})T$ and variance $\sigma^2 T$.

Page 54: Black-Scholes Put Formula

Consider a European put on the stock expiring in T years and having strike price K .

We want to compute

$$P_0 = \lim_{N \rightarrow \infty} \left[\frac{1}{\left(1 + \frac{rT}{N}\right)^N} \tilde{\mathbb{E}}^{(N)}[(K - S_N^{(N)})^+] \right].$$

Using a central-limit argument, it can be shown that

$$P_0 = e^{-rT} \int_{-\infty}^{\infty} (K - e^x)^+ \psi(x) dx,$$

where ψ is the density function of a normal random variable with mean

$$\mu = \ln S_0 + \left(r - \frac{1}{2}\sigma^2\right)T$$

and standard deviation

$$\nu = \sigma\sqrt{T} \quad (\text{variance } \nu^2).$$

Page 55: Black-Scholes Put Formula (Cont.)

Recall that

$$\psi(x) = \frac{1}{\nu} \phi\left(\frac{x - \mu}{\nu}\right),$$

where ϕ is the standard normal density function, i.e.

$$\phi(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}.$$

Making the change of variable

$$y = \frac{x - \mu}{\nu},$$

we find that

$$P_0 = e^{-rT} \int_{-\infty}^{\infty} (K - e^{\mu + \nu y})^+ \phi(y) dy.$$

Page 56: Black-Scholes Put Formula (Cont.)

Notice that the integrand vanishes identically when

$$\mu + \nu y \geq \ln K,$$

i.e. when

$$y \geq \frac{\ln K - \mu}{\nu}$$

If we put

$$c = \frac{\ln K - \mu}{\nu},$$

we find that

$$P_0 = e^{-rT} \int_{-\infty}^c (K - e^{\mu + \nu y}) \phi(y) dy.$$

Page 57: Black-Scholes Put Formula (Cont.)

Let us set

$$I = \int_{-\infty}^c e^{\mu + \nu y} \phi(y) dy.$$

Then the formula for P_0 reduces to

$$P_0 = e^{-rT} K \Phi(c) - e^{-rT} I,$$

where Φ is the cumulative distribution function for a standard normal random variable. Notice that

$$e^{\mu + \nu y} \phi(y) = \frac{e^{\mu}}{\sqrt{2\pi}} e^{\nu y - \frac{1}{2}y^2}.$$

Completing the square, we see that

$$\nu y - \frac{1}{2}y^2 = -\frac{1}{2}(y - \nu)^2 + \frac{1}{2}\nu^2,$$

so that

$$I = \frac{e^{\mu + \frac{1}{2}\nu^2}}{\sqrt{2\pi}} \int_{-\infty}^c \exp \left\{ -\frac{1}{2}(y - \sigma\sqrt{T})^2 \right\} dy.$$

Page 58: Black-Scholes Put Formula (Cont.)

Making the change of variable $x = y - \nu$, we find that

$$I = e^{\mu + \frac{1}{2}\nu^2} \int_{-\infty}^{c-\nu} \phi(x) dx = e^{\mu + \frac{1}{2}\nu^2} \Phi(c - \nu).$$

Finally, we arrive at

$$P_0 = e^{-rT} K \Phi(c) - e^{-rT} e^{\mu + \frac{1}{2}\nu^2} \Phi(c - \nu).$$

Using the definitions of μ and ν from page 54 we obtain

$$-rT + \mu + \frac{1}{2}\nu^2 = \ln S_0.$$

It follows that

$$P_0 = Ke^{-rT} \Phi(c) - S_0 \Phi(c - \nu).$$

Page 59: Black-Scholes Put Formula (Cont.)

It is common practice to define

$$d_1(T, S_0) = \frac{1}{\sigma\sqrt{T}} \left[\ln \left(\frac{S_0}{K} \right) + \left(r + \frac{\sigma^2}{2} \right) T \right] = -c + \nu,$$

$$d_2(T, S_0) = \frac{1}{\sigma\sqrt{T}} \left[\ln \left(\frac{S_0}{K} \right) + \left(r - \frac{\sigma^2}{2} \right) T \right] = -c,$$

so that

$$P_0 = e^{-rT} K \Phi(-d_2(T, S_0)) - S_0 \Phi(-d_1(T, S_0)).$$

Page 60: Black-Scholes Call Formula

We can use put-call parity to obtain the price C_0 of a European call on the stock expiring at time T and having strike price K :

$$\begin{aligned}C_0 &= P_0 + S_0 - Ke^{-rT} \\&= S_0(1 - \Phi(-d_1(T, S_0))) + Ke^{-rT}(\Phi(-d_2(T, S_0)) - 1).\end{aligned}$$

Using the fact that $1 - \Phi(-x) = \Phi(x)$ we obtain

$$C_0 = S_0\Phi(d_1(T, S_0)) - Ke^{-rT}\Phi(d_2(T, S_0)).$$

Page 61: Implied Volatility

Exchange-traded puts and calls on stocks are typically American, rather than European. However, when there are no dividends and $r \geq 0$, an American call on a given stock has the same price as the European call with the same strike and expiration. For a given call option, there is a “magic” value of σ for which the Black-Scholes price agrees with the market price; this value of σ is called the **implied volatility** of the option.

If the Black-Scholes theory worked perfectly, then for a given maturity, the implied volatility would be the same for all strike prices. However, in practice, when the implied volatility is plotted against strike, the graph typically looks like a “smile” or “smirk”, called a **volatility smile** or **volatility smirk**.

Page 62: Other Possible Choices of $u^{(N)}$, $d^{(N)}$, and $r^{(N)}$

Cox, Ross, and Rubinstein used

$$u^{(N)} = e^{\sigma\sqrt{\frac{T}{N}}}, \quad d^{(N)} = e^{-\sigma\sqrt{\frac{T}{N}}}, \quad r^{(N)} = e^{\frac{rT}{N}} - 1.$$

The corresponding one-step risk-neutral probabilities are given by

$$\tilde{p}^{(N)} = \frac{e^{\frac{rT}{N}} - e^{-\sigma\sqrt{\frac{T}{N}}}}{e^{\sigma\sqrt{\frac{T}{N}}} - e^{-\sigma\sqrt{\frac{T}{N}}}}, \quad \tilde{q}^{(N)} = 1 - \tilde{p}^{(N)}.$$

Shreve (Unpublished Chapter 8 to Vol I) used

$$u^{(N)} = 1 + \frac{rT}{N} + \frac{\sigma\sqrt{T}}{\sqrt{N}}, \quad d^{(N)} = 1 + \frac{rT}{N} - \frac{\sigma\sqrt{T}}{\sqrt{N}},$$
$$r^{(N)} = \frac{rT}{N}.$$

The corresponding one-step risk-neutral probabilities are

$$\tilde{p}^{(N)} = \frac{1}{2}, \quad \tilde{q}^{(N)} = \frac{1}{2}.$$

Page 63: Option Prices as Time Evolves

Here we consider European calls and puts expiring at time $T > 0$. We look at the prices $c(t, x)$ and $p(t, x)$ at time $t \in [0, T]$, given that the price of the stock at time t is x . Notice that the time to maturity is $T - t$. (Some folks denote the time to maturity by the single letter τ .) Derivatives with respect to t are the “negatives” of derivative with respect to τ . We have

$$c(t, x) = x\Phi(d_1(T - t, x)) - Ke^{-r(T-t)}\Phi(d_2(T - t, x)),$$

$$p(t, x) = -x\Phi(-d_1(T - t, x)) + Ke^{-r(T-t)}\Phi(-d_2(T - t, x)).$$

Page 64: Black-Scholes PDE

Let $v(t, x)$ denote the price at time t of a European-style derivative security (with maturity T) given that the price of the stock at time t is x . (Here $v(t, x)$ could be either $c(t, x)$ or $p(t, x)$ or the price of some other option.) v satisfies the Black-Scholes PDE:

$$v_t(t, x) + rxv_x(t, x) + \frac{1}{2}\sigma^2x^2v_{xx}(t, x) - rv(t, x) = 0.$$

One can solve the PDE, together with appropriate auxiliary conditions, to determine option prices.

Page 65: The Greeks

Changes in option prices with respect to changes in various parameters (with all other parameters held fixed) are known as “Greeks”. There are many Greeks for options. For now, I will introduce the 5 most basic: Δ (Delta), Γ (Gamma), Θ (Theta), $\mathcal{V}$ (Vega), and ρ (Rho):

$$\Delta = v_x(t, x),$$

$$\Gamma = v_{xx}(t, x),$$

$$\Theta = v_t(t, x),$$

$$\mathcal{V} = \frac{\partial}{\partial \sigma} v(t, x),$$

$$\rho = \frac{\partial}{\partial r} v(t, x).$$

Page 66: The Greeks (Cont.)

- ▶ $\Delta > 0$ for a call and $\Delta < 0$ for a put.
- ▶ $\Gamma > 0$ both for calls and puts.
- ▶ $\Theta < 0$ for calls and

$$\Theta(\text{put}) = \Theta(\text{call}) + rKe^{-r(T-t)}.$$

Θ is usually negative for a put, but a deep-in-the-money put can have $\Theta > 0$.

- ▶ $\mathcal{V} > 0$ for both calls and puts.
- ▶ ρ is positive for calls and negative for puts.

Page 67: Greeks for a Call

Exercise 1: Show that

$$\Delta = \Phi(d_1(T - t, x)),$$

$$\Gamma = \frac{\Phi'(d_1(T - t, x))}{x\sigma\sqrt{T - t}},$$

$$\Theta = -rKe^{-r(T-t)}\Phi(d_2(T - t, x)) - \frac{\sigma x}{2\sqrt{T - t}}\Phi'(d_1(T - t, x)),$$

$$\mathcal{V} = x\sqrt{T - t}\Phi'(d_1(T - t, x)),$$

$$\rho = K(T - t)e^{-r(T-t)}N(d_2).$$

Page 68: Greeks for a Put

Exercise 2: Use put-call parity

$$p(t, x) = c(t, x) + Ke^{-r(T-t)} - x,$$

to compute the Greeks for a put option.

We can think of modeling the stock price at other times $t \in (0, T]$ by taking suitable limit of stock prices in the binomial model. We can also model the evolution of the stock price in continuous time directly with the stochastic differential equations

$$dS(t) = S(t)(\alpha dt + \sigma dW(t)),$$

$$dS(t) = S(t)(r dt + \sigma d\tilde{W}(t)),$$

where $(W(t))_{t \geq 0}$ is a Brownian motion under the actual measure and $(\tilde{W}(t))_{t \geq 0}$ is a Brownian motion under the risk-neutral measure.

Page 70: Stock Price Paths (Cont.)

Using Itô's Lemma, the solutions of these equations can be written as

$$\ln \left(\frac{S(t)}{S(0)} \right) = \left(\alpha - \frac{1}{2} \sigma^2 \right) t + \sigma W(t),$$

and

$$\ln \left(\frac{S(t)}{S(0)} \right) = \left(r - \frac{1}{2} \sigma^2 \right) t + \sigma \tilde{W}(t).$$

You will do this carefully in Stochastic Calculus.

Page 71: Stock Price Path (Cont.)

Now let us divide the time interval $[0, T]$ into n equal steps of length

$$\Delta T = \frac{T}{n}$$

and set

$$t_i = i\Delta T, \quad i = 0, 1, \dots, n.$$

Then, under the risk-neutral measure the random variables

$$\ln \left(\frac{S(t_i + \Delta T)}{S(t_i)} \right), \quad i = 0, 1, \dots, n-1$$

are independent identically distributed normal with mean $(r - \frac{\sigma^2}{2})\Delta T$ and variance $\sigma^2\Delta T$.

Page 72: Stock Path Simulation Under $\tilde{\mathbb{P}}$

If Z is standard normal, then the distribution of

$$\left(r - \frac{\sigma^2}{2}\right)\Delta T + \sigma\sqrt{\Delta T}Z$$

is normal with mean $\left(r - \frac{\sigma^2}{2}\right)\Delta T$ and variance $\sigma^2\Delta T$.

We can simulate stock price paths under the risk-neutral measure by simulating independent standard normal random variables

Z_i , $i = 0, 1, \dots, n-1$ and putting

$$S(t_i + \Delta T) = S(t_i) \exp[a + bZ_i],$$

where

$$a = \left(r - \frac{\sigma^2}{2}\right)\Delta T, \quad b = \sigma\sqrt{\Delta T}.$$

Page 73: Historical Volatility

Under the actual measure the random variables

$$Y_i = \ln \left(\frac{S(t_i + \Delta T)}{S(t_i)} \right), \quad i = 0, 1, \dots, n-1$$

are independent identically distributed normal with mean $(\alpha - \frac{\sigma^2}{2})\Delta T$ and variance $\sigma^2\Delta T$.

We could try to estimate α and σ by using historical stock prices. Assuming that the model is correct, we can estimate σ as accurately as we want by sampling frequently (i.e. using small ΔT). Such an estimate for σ is called **historical volatility**.

Page 74: Can We Estimate α ?

However, it is impossible to obtain a useful estimate for α from historical stock prices. Frequent sampling does not help; we would need to have a very long time interval.

To see why, let us assume that σ is known exactly and put

$$Y_i = \ln \left(\frac{S(t_i + \Delta T)}{S(t_i)} \right), i = 0, 1, 2, \dots, n-1.$$

A natural estimator for α is

$$\hat{\alpha} = \frac{1}{2}\sigma^2 + \frac{1}{n\Delta T} \sum_{i=0}^{n-1} Y_i.$$

Notice that the sample mean

$$\frac{1}{n} \sum_{i=0}^{n-1} Y_i$$

is normal with mean $(\alpha - \frac{\sigma^2}{2})\Delta T$ and variance $(\sigma^2\Delta T)/n$

Page 75: Confidence Interval for α

The Lazy Mathematician's 95% Confidence Interval: If $X \sim \mathcal{N}(\mu, \sigma^2)$ then

$$\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx .95.$$

Let $\delta > 0$ be given. Suppose that we want

$$|\hat{\alpha} - \alpha| < \delta,$$

with probability .95 or greater. Then we would need

$$T \geq \frac{4\sigma^2}{\delta^2}.$$

With $\sigma = .2$ (a very reasonable value) and $\delta = .01$, we would need $T=1,600$ years. If we use approximately 10 years of data for a typically stock, a 95% confidence interval for α would be something like $[0, .25]$. Such an estimate is useless!

Page 76: Estimating α (Cont.)

To help understand what is going on here, it is useful to observe that

$$\sum_{i=0}^{n-1} Y_i = \ln \left(\frac{S(T)}{S(0)} \right).$$

This shows that the estimator for α depends on the observations only through the ratio of the terminal stock price to the initial stock price.

The classical Black-Scholes Theory assumes that traders can borrow or invest at a risk-free rate r . This ignores the intricacies of collateral and multiple interest rates. Collateral has become increasingly important since the 2008 financial crisis. In 2010 Vladimir Piterbarg developed a framework to account for collateral and multiple interest rates in Black Scholes. These factors necessitate changes to discounting, forward prices and implied volatilities. You will study this in Stochastic Calculus II.